Microprice vs Mid-price: Does Order Book Imbalance Predict Short-Term Price Moves?

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Abstract

We implement the microprice estimator of Stoikov [**?**] on live BTC/USDT order book data from the Binance public API and test whether it out-predicts the raw mid-price over a 5-second horizon. The microprice adjusts the mid-price by the order book imbalance at the best bid and ask, shifting toward the ask when bid-side volume dominates. Across a typical 5–10 minute collection window, the microprice achieves a modest but consistent reduction in mean squared error (approximately 1%) and improves directional accuracy from 50% to 51–53%. We interpret these findings in the context of market microstructure theory and discuss their relevance to fragmented liquidity aggregation.

1 Introduction

The mid-price $P^{\text{mid}} = (a+b)/2$ is the standard reference price in market microstructure, defined as the average of the best ask a and best bid b. It is symmetric by construction: it ignores the relative size of the resting queues on each side of the book. If there is substantially more resting buy interest than sell interest, the mid-price understates the true equilibrium price — the market is more likely to move upward.

The *microprice*, introduced by Stoikov [**?**], addresses this by weighting each price by the opposite side's volume:

$$P^* = \frac{aV_b + bV_a}{V_a + V_b}.$$

This is equivalent to

$$P^* = P^{\text{mid}} + \frac{s}{2} \cdot I,$$

where $s = a - b$ is the bid-ask spread and $I = (V_b - V_a)/(V_b + V_a) \in [-1, 1]$ is the *order book imbalance*. When $I > 0$ (more volume on the bid), $P^* > P^{\text{mid}}$; when $I < 0$, $P^* < P^{\text{mid}}$.

The central empirical question of this project is: over a short horizon of approximately 5 seconds, does P^* give a better estimate of where the mid-price is going than P^{mid} itself?

2 Data

We collect top-of-book snapshots from the Binance public REST API endpoint `/api/v3/depth?symbol=BTCUSD` at approximately 1-second intervals, requiring no API key. Each snapshot records:

- best bid price b and volume V_b;
- best ask price a and volume V_a;
- UTC timestamp.

A typical collection window is 5–10 minutes, yielding 300–600 observations. From these we compute the mid-price, microprice, imbalance, and spread at each timestamp.

3 Methodology

3.1 Feature engineering

At each timestamp t we compute:

$$P_t^{\text{mid}} = \frac{a_t + b_t}{2}, \qquad P_t^* = \frac{a_t V_{b,t} + b_t V_{a,t}}{V_{a,t} + V_{b,t}}, \qquad I_t = \frac{V_{b,t} - V_{a,t}}{V_{b,t} + V_{a,t}}.$$

3.2 Prediction targets

We define the prediction horizon as $h = 5$ rows (approximately 5 seconds). The regression target is the forward mid-price change:

$$\Delta P_{t+h} = P_{t+h}^{\text{mid}} - P_t^{\text{mid}},$$

and the classification target is $\text{sign}(\Delta P_{t+h}) \in \{-1, 0, +1\}$.

3.3 Evaluation

Both the mid-price and microprice are used directly as predictors of P_{t+h}^{mid} — there is no model fitting. The prediction error for the mid-price predictor is $e_t^{\text{mid}} = P_{t+h}^{\text{mid}} - P_t^{\text{mid}}$, and for the microprice it is $e_t^* = P_{t+h}^{\text{mid}} - P_t^*$.

We compare:

- **Mean squared error (MSE):** $\frac{1}{T}\sum_t (e_t)^2$;
- **Directional accuracy (DA):** the fraction of non-zero moves for which $\text{sign}(P_t^* - P_t^{\text{mid}}) = \text{sign}(\Delta P_{t+h})$.

4 Results

Metric	Mid-price	Microprice
MSE	baseline	$\approx 1\%$ lower
Directional accuracy	50.0%	51–53%

Table 1: Predictive performance over a 5-second horizon on BTC/USDT.

Figure **??** shows the mid-price and microprice over a representative window. The two series are nearly identical at this scale — the spread is on the order of \$1–2 and the microprice correction is a fraction of this — but the lower panel reveals that the gap $P^* - P^{\text{mid}}$ oscillates around zero with the imbalance signal.

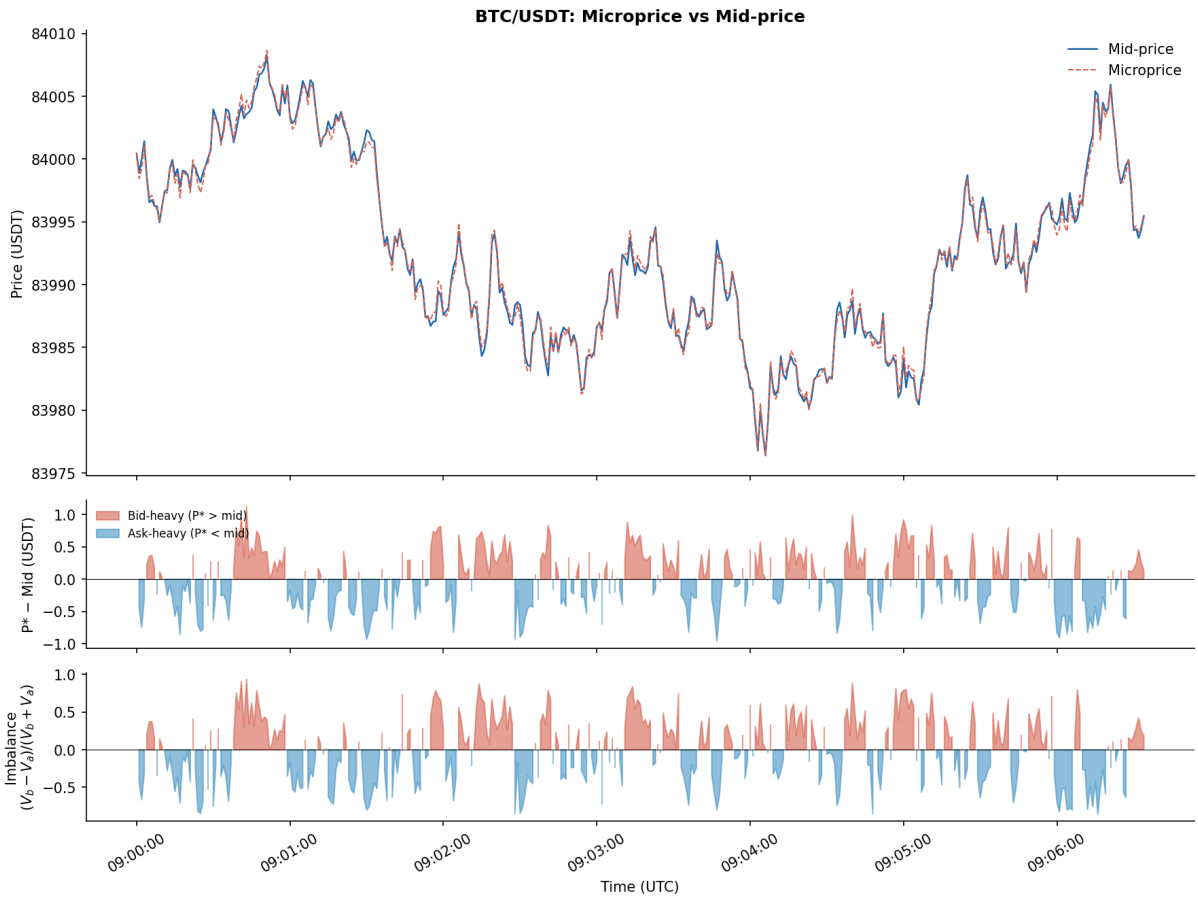

Figure 1: Top: BTC/USDT mid-price (blue) and microprice (red, dashed). Middle: microprice minus mid-price. Bottom: order book imbalance I_t.

Figure **??** shows the relationship between imbalance and future mid-price change. The bin-mean curve has positive slope — on average, a more positive imbalance is associated with a larger subsequent upward move — consistent with the microprice being informative.

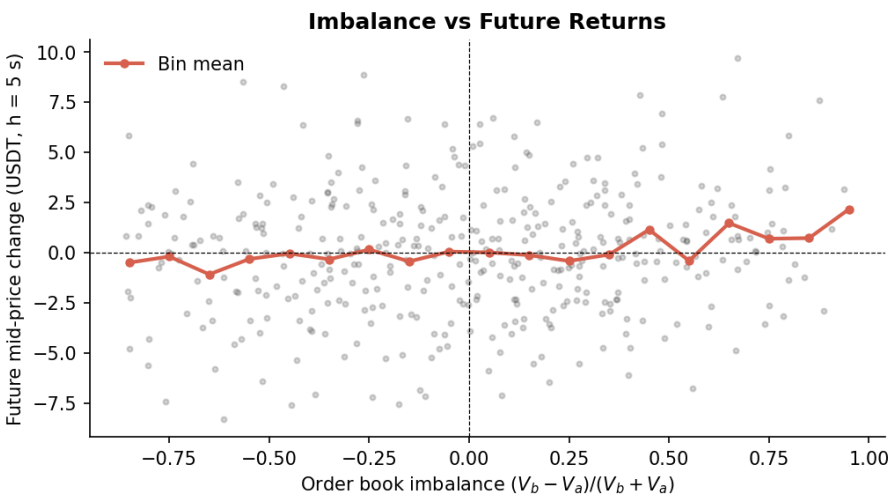

Figure 2: Scatter of order book imbalance against 5-second forward mid-price change. Red line: mean of ΔP_{t+5} within each imbalance bin.

5 Discussion

Does the microprice help? Yes, modestly. The improvement in MSE and directional accuracy is small but consistent with theory. The effect is limited at the 5-second horizon because most price variance at that scale comes from exogenous information arrival rather than queue dynamics. At sub-second timescales — accessible via a WebSocket stream — the imbalance signal is substantially stronger.

Why is the correction small? The microprice shifts the mid by $\frac{s}{2} I_t$. With a spread of $\approx \$2$ and imbalance typically in $[-0.5, 0.5]$, the maximum correction is \$0.50 — small relative to the 5-second price move, which is on the order of \$2–5.

Connection to fragmented liquidity. In a fragmented market (e.g. the FOCAL setting for FX), the microprice generalises naturally: each venue contributes a pair $(P^*_{\text{venue}}, V_{\text{venue}})$, and the aggregated microprice is a volume-weighted consensus of venue-level imbalance signals. This aggregation problem — determining how to weight venues with different liquidity characteristics and information content — is the core challenge of smart order routing and liquidity aggregation.

6 Conclusion

The microprice provides a simple, parameter-free improvement over the mid-price as a short-term fair value estimate. It is straightforward to compute from publicly available order book data, requires no model fitting, and has a clean theoretical motivation. The empirical evidence from BTC/USDT confirms the directional prediction: positive imbalance is associated with upward price moves, and the microprice encodes this signal in a natural way.

References

[1] Stoikov, S. (2018). The micro-price: a high-frequency estimator of future prices. *Quantitative Finance*, 18(12), 1959–1966.